	School of Engineering & Technology	
	Department: SOET	Session: 25-26 Even
	Program: B. Tech CSE	Semester: IV
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	Course Name: Database Management Systems Lab	Faculty: Ms. Mansi kajal

Lab Sheet: 2

Objective:

• To understand and implement SQL transactions using COMMIT, SAVEPOINT, and ROLLBACK, retrieve records using SELECT queries, and perform INSERT, UPDATE, and ALTER operations on tables.

Instructions:

1. Open your preferred Database Management System (DBMS) tool.
2. Create the required tables: CLASS, EMPLOYEES, and Client_master.
3. Insert the given/suitable records into each table.
4. Execute the SQL scripts and queries step by step.
5. Observe the output carefully after each operation, especially after ROLLBACK and UPDATE commands.

Table Structures:

CLASS

D (NUMBER) and Name (VARCHAR).

The initial records in the table are (1, 'Bravo'), (2, 'Alex'), and (4, 'Cheng').

DDL Questions (Data Definition Language):

1. How do you create a table named CLASS with appropriate data types?

```
mysql> create database DBMS_Vandana;
Query OK, 1 row affected (0.01 sec)

mysql> use DBMS_Vandana;
Database changed
mysql> CREATE TABLE CLASS (
    ->     ID INT PRIMARY KEY,
    ->     Name VARCHAR(40)
    -> );
Query OK, 0 rows affected (0.03 sec)
```

2. How do you create a table named EMPLOYEES with the specified attributes?

```
mysql>  use DBMS_Vandana;
Database changed
mysql> CREATE TABLE EMPLOYEES (
    ->     Employee_Id INT,
    ->     First_Name VARCHAR(20),
    ->     Last_Name VARCHAR(20),
    ->     Email VARCHAR(30),
    ->     Phone_Number VARCHAR(15),
    ->     Hire_Date DATE,
    ->     Job_Id VARCHAR(20),
    ->     Salary INT,
    ->     Commission_Pct DECIMAL(5,2),
    ->     Manager_Id INT,
    ->     Department_Id INT
    -> );
Query OK, 0 rows affected (0.02 sec)
```

3. How do you create a table named Client_master with the specified attributes?

```
mysql> CREATE TABLE Client_master (
    ->     ClientNo VARCHAR(10) PRIMARY KEY,
    ->     Name VARCHAR(30),
    ->     Address VARCHAR(50),
    ->     City VARCHAR(20),
    ->     State VARCHAR(20),
    ->     Pincode INT,
    ->     Bal_Due INT
    -> );
Query OK, 0 rows affected (0.02 sec)
```

4.What SQL statement would you use to rename the table Client_master to Client12?
Ans: RENAME TABLE Client_master TO Client12;

5.How do you display the bal_due column heading as “BALANCE”?
Ans: SELECT Bal_Due AS BALANCE
FROM Client_master;

DML Questions (Data Manipulation Language):

Part A: Transactions and Savepoints (CLASS Table)
Give the output of the following SQL script:

- INSERT INTO class VALUES (5,’Rahul’);
- COMMIT;

- UPDATE class SET name = ‘Abhijeet’ WHERE id=’5’;
- SAVEPOINT A;
- INSERT INTO class VALUES (6,’Chris’);
- SAVEPOINT B;
- INSERT INTO class VALUES (7,’Bravo’);
- SAVEPOINT C;
- SELECT * FROM class;
- ROLLBACK TO B;
- SELECT * FROM class;
- ROLLBACK TO A;

```
mysql> create database DBMS_Vandana;
Query OK, 1 row affected (0.01 sec)

mysql> use DBMS_Vandana;
Database changed
mysql> CREATE TABLE CLASS (
    ->     ID INT PRIMARY KEY,
    ->     Name VARCHAR(40)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO CLASS VALUES
    -> (1,'Bravo'),
    -> (2,'Alex'),
    -> (4,'Cheng');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM CLASS;
+----+-------+
| ID | Name  |
+----+-------+
|  1 | Bravo |
|  2 | Alex  |
|  4 | Cheng |
+----+-------+
3 rows in set (0.00 sec)

mysql> SET autocommit = 0;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO CLASS VALUES (5,'Rahul');
Query OK, 1 row affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> UPDATE CLASS SET Name='Abhijeet' WHERE ID=5;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SAVEPOINT A;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> ROLLBACK TO A;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM CLASS;
+----+----------+
| ID | Name     |
+----+----------+
|  1 | Bravo    |
|  2 | Alex     |
|  4 | Cheng    |
|  5 | Abhijeet |
+----+----------+
4 rows in set (0.00 sec)

mysql> SELECT * FROM CLASS;
+----+----------+
| ID | Name     |
+----+----------+
|  1 | Bravo    |
|  2 | Alex     |
|  4 | Cheng    |
|  5 | Abhijeet |
+----+----------+
4 rows in set (0.00 sec)
```

Part B: Retrieving Records (EMPLOYEES Table)

1. Find out the employee id, names, and salaries of all the employees.
2. List out the employees who work under manager 100.
3. Find the names of the employees who have a salary greater than or equal to 4800.
4. List out the employees whose last name is 'AUSTIN'.
5. Find the names of the employees who work in departments 60, 70, and 80.
6. Display the unique Manager_Id.

```
mysql>  use DBMS_Vandana;
Database changed
mysql> CREATE TABLE EMPLOYEES (
    ->     Employee_Id INT,
    ->     First_Name VARCHAR(20),
    ->     Last_Name VARCHAR(20),
    ->     Email VARCHAR(30),
    ->     Phone_Number VARCHAR(15),
    ->     Hire_Date DATE,
    ->     Job_Id VARCHAR(20),
    ->     Salary INT,
    ->     Commission_Pct DECIMAL(5,2),
    ->     Manager_Id INT,
    ->     Department_Id INT
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO EMPLOYEES VALUES
    -> (101,'John','Austin','john@gmail.com','9999999999','2020-01-10','IT_PROG',5000,NULL,100,60),
    -> (102,'Sara','Smith','sara@gmail.com','8888888888','2019-03-15','HR_REP',4500,NULL,100,70),
    -> (103,'Mike','Brown','mike@gmail.com','7777777777','2018-06-20','SA_REP',6000,NULL,101,80),
    -> (104,'David','Austin','david@gmail.com','6666666666','2021-07-11','IT_PROG',4800,NULL,100,60),
    -> (105,'Neha','Khan','neha@gmail.com','5555555555','2022-02-01','HR_REP',7000,NULL,102,70);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM EMPLOYEES;
+-------------+------------+-----------+----------------+--------------+------------+---------+--------+----------------+------------+---------------+
| Employee_Id | First_Name | Last_Name | Email          | Phone_Number | Hire_Date  | Job_Id  | Salary | Commission_Pct | Manager_Id | Department_Id |
+-------------+------------+-----------+----------------+--------------+------------+---------+--------+----------------+------------+---------------+
|         101 | John       | Austin    | john@gmail.com | 9999999999   | 2020-01-10 | IT_PROG |   5000 |           NULL |        100 |            60 |
|         102 | Sara       | Smith     | sara@gmail.com | 8888888888   | 2019-03-15 | HR_REP  |   4500 |           NULL |        100 |            70 |
|         103 | Mike       | Brown     | mike@gmail.com | 7777777777   | 2018-06-20 | SA_REP  |   6000 |           NULL |        101 |            80 |
|         104 | David      | Austin    | david@gmail.com | 6666666666   | 2021-07-11 | IT_PROG |   4800 |           NULL |        100 |            60 |
|         105 | Neha       | Khan      | neha@gmail.com | 5555555555   | 2022-02-01 | HR_REP  |   7000 |           NULL |        102 |            70 |
+-------------+------------+-----------+----------------+--------------+------------+---------+--------+----------------+------------+---------------+
5 rows in set (0.00 sec)

mysql> SELECT Employee_Id, First_Name, Last_Name, Salary
    -> FROM EMPLOYEES;
+-------------+------------+-----------+--------+
| Employee_Id | First_Name | Last_Name | Salary |
+-------------+------------+-----------+--------+
|         101 | John       | Austin    |   5000 |
|         102 | Sara       | Smith     |   4500 |
```

```
| Employee_Id | First_Name | Last_Name | Salary |
+-------------+------------+-----------+--------+
|         101 | John       | Austin    |   5000 |
|         102 | Sara       | Smith     |   4500 |
|         103 | Mike       | Brown     |   6000 |
|         104 | David      | Austin    |   4800 |
|         105 | Neha       | Khan      |   7000 |
+-------------+------------+-----------+--------+
5 rows in set (0.00 sec)

mysql> SELECT *
    -> FROM EMPLOYEES
    -> WHERE Manager_Id = 100;
+-------------+------------+-----------+-----------------+--------------+------------+---------+--------+----------------+------------+---------------+
| Employee_Id | First_Name | Last_Name | Email           | Phone_Number | Hire_Date  | Job_Id  | Salary | Commission_Pct | Manager_Id | Department_Id |
+-------------+------------+-----------+-----------------+--------------+------------+---------+--------+----------------+------------+---------------+
|         101 | John       | Austin    | john@gmail.com  | 9999999999   | 2020-01-10 | IT_PROG |   5000 |           NULL |        100 |            60 |
|         102 | Sara       | Smith     | sara@gmail.com  | 8888888888   | 2019-03-15 | HR_REP  |   4500 |           NULL |        100 |            70 |
|         104 | David      | Austin    | david@gmail.com | 6666666666   | 2021-07-11 | IT_PROG |   4800 |           NULL |        100 |            60 |
+-------------+------------+-----------+-----------------+--------------+------------+---------+--------+----------------+------------+---------------+
3 rows in set (0.00 sec)

mysql> SELECT First_Name, Last_Name
    -> FROM EMPLOYEES
    -> WHERE Salary >= 4800;
+------------+-----------+
| First_Name | Last_Name |
+------------+-----------+
| John       | Austin    |
| Mike       | Brown     |
| David      | Austin    |
| Neha       | Khan      |
+------------+-----------+
4 rows in set (0.00 sec)

mysql> SELECT *
    -> FROM EMPLOYEES
    -> WHERE Last_Name = 'Austin';
+-------------+------------+-----------+-----------------+--------------+------------+---------+--------+----------------+------------+---------------+
| Employee_Id | First_Name | Last_Name | Email           | Phone_Number | Hire_Date  | Job_Id  | Salary | Commission_Pct | Manager_Id | Department_Id |
+-------------+------------+-----------+-----------------+--------------+------------+---------+--------+----------------+------------+---------------+
|         101 | John       | Austin    | john@gmail.com  | 9999999999   | 2020-01-10 | IT_PROG |   5000 |           NULL |        100 |            60 |
|         104 | David      | Austin    | david@gmail.com | 6666666666   | 2021-07-11 | IT_PROG |   4800 |           NULL |        100 |            60 |
+-------------+------------+-----------+-----------------+--------------+------------+---------+--------+----------------+------------+---------------+
2 rows in set (0.00 sec)
```

```
ysql> SELECT First_Name, Last_Name
    -> FROM EMPLOYEES
    -> WHERE Department_Id IN (60,70,80);
------------+-----------+
 First_Name | Last_Name |
------------+-----------+
 John       | Austin    |
 Sara       | Smith     |
 Mike       | Brown     |
 David      | Austin    |
 Neha       | Khan      |
------------+-----------+
 rows in set (0.00 sec)

ysql> SELECT DISTINCT Manager_Id
    -> FROM EMPLOYEES;
------------+
 Manager_Id |
------------+
        100 |
        101 |
        102 |
------------+
 rows in set (0.00 sec)
```

Part C: Updating Records (Client_master Table)

1. Insert five records into the Client_master table.
2. Find the names of clients whose bal_due > 5000.
3. Change the bal_due of ClientNO "C123" to Rs. 5100

```
mysql> CREATE TABLE Client_master (
    ->     ClientNo VARCHAR(10) PRIMARY KEY,
    ->     Name VARCHAR(30),
    ->     Address VARCHAR(50),
    ->     City VARCHAR(20),
    ->     State VARCHAR(20),
    ->     Pincode INT,
    ->     Bal_Due INT
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO Client_master VALUES
    -> ('C123','Raj','Delhi Street','Delhi','Delhi',110001,6000),
    -> ('C124','Amit','Link Road','Mumbai','MH',400001,3000),
    -> ('C125','Sara','Anna Nagar','Chennai','TN',600001,7000),
    -> ('C126','John','Park Street','Kolkata','WB',700001,2000),
    -> ('C127','Neha','MG Road','Pune','MH',411001,8000);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM Client_master;
+----------+------+--------------+---------+-------+---------+---------+
| ClientNo | Name | Address      | City    | State | Pincode | Bal_Due |
+----------+------+--------------+---------+-------+---------+---------+
| C123     | Raj  | Delhi Street | Delhi   | Delhi |  110001 |    6000 |
| C124     | Amit | Link Road    | Mumbai  | MH    |  400001 |    3000 |
| C125     | Sara | Anna Nagar   | Chennai | TN    |  600001 |    7000 |
| C126     | John | Park Street  | Kolkata | WB    |  700001 |    2000 |
| C127     | Neha | MG Road      | Pune    | MH    |  411001 |    8000 |
+----------+------+--------------+---------+-------+---------+---------+
5 rows in set (0.00 sec)

mysql> SELECT Name
    -> FROM Client_master
    -> WHERE Bal_Due > 5000;
+------+
| Name |
+------+
| Raj  |
| Sara |
| Neha |
+------+
3 rows in set (0.00 sec)
```

```
mysql> UPDATE Client_master
    -> SET Bal_Due = 5100
    -> WHERE ClientNo = 'C123';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM Client_master;
+----------+------+--------------+---------+-------+---------+---------+
| ClientNo | Name | Address      | City    | State | Pincode | Bal_Due |
+----------+------+--------------+---------+-------+---------+---------+
| C123     | Raj  | Delhi Street | Delhi   | Delhi |  110001 |    5100 |
| C124     | Amit | Link Road    | Mumbai  | MH    |  400001 |    3000 |
| C125     | Sara | Anna Nagar   | Chennai | TN    |  600001 |    7000 |
| C126     | John | Park Street  | Kolkata | WB    |  700001 |    2000 |
| C127     | Neha | MG Road      | Pune    | MH    |  411001 |    8000 |
+----------+------+--------------+---------+-------+---------+---------+
5 rows in set (0.00 sec)

mysql>
```